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MATHEMATICS

TARGET : JEE (Main) 2024

CHAMP-2024 REVISION DPP No. 25

Syllabus :- Probability

SECTION - I

Multiple Choice Questions (MCQ)

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

1. If A and B are two events such that P(1) > 0 and P(2) ≠ 1 then $P(\bar{A}/\bar{B})$ is equal to-

(1) $1 - P\left(\frac{A}{B}\right)$ (2) $1 - P\left(\frac{\bar{A}}{B}\right)$ (3) $\frac{1-P(A\cup B)}{P(\bar{B})}$ (4) $\frac{P(\bar{A})}{P(\bar{B})}$

2. A determinant of second order is made with the elements 0 and 1. What is the probability that the value of determinant is non-negative?

(1) $\frac{13}{16}$ (2) $\frac{3}{16}$ (3) $\frac{3}{4}$ (4) $\frac{7}{8}$

3. If the letters of the word **"ATTEMPT"** are written down at random then the probability that all the T's are consecutive is

(1) $\frac{1}{42}$ (2) $\frac{6}{7}$ (3) $\frac{1}{7}$ (4) $\frac{1}{6}$

4. A dice is rolled three times. Let E_1 denotes the event of getting a number larger than the previous number each time and E_2 denotes the event that the numbers (in order) form an increasing AP then –

(1) $P(E_2) \geq P(E_1)$ (2) $P(E_2 \cap E_1) = 3/10$ (3) $P(E_2/E_1) = 1/36$ (4) $P(E_1) = \frac{10}{3} P(E_2)$

5. Let the 9 different letters A, B, C.... I ∈ {1, 2, 3,....9} then the probability that product (A – 1)(B – 1)..... (I – 9) is even number will be-

(1) $\frac{1}{2}$ (2) $\frac{1}{4}$ (3) 1 (4) 0

6. The probability that A speaks truth is $\frac{4}{5}$ while this probability for B is $\frac{3}{4}$. The probability that they contradict each other when asked to speak on a fact, is

(1) $\frac{3}{20}$ (2) $\frac{1}{5}$ (3) $\frac{7}{20}$ (4) $\frac{4}{5}$

7. In a certain college, 25% of the students failed mathematics 15% failed chemistry and 10% failed both mathematics and chemistry. A student is selected at random. If the student failed chemistry then find probability that the student failed mathematics also.

(1) $\frac{2}{9}$ (2) $\frac{2}{3}$ (3) $\frac{1}{3}$ (4) $\frac{1}{9}$

8. From a group of 10 persons consisting of 5 lawyers, 3 doctors and 2 engineers, four persons are selected at random. The probability that the selection contains at least one of each category, is

(1) $\frac{1}{2}$ (2) $\frac{1}{3}$ (3) $\frac{2}{3}$ (4) $\frac{2}{9}$

9. In a lot of screws, 55% are produced by machine A and the rest by machine B. Machine A produces 2% defective screws and machine B produces 3% defective screws. Find probability that a defective screw randomly found from the lot is manufactured by machine B is

(1) $\frac{3}{5}$ (2) $\frac{29}{51}$ (3) $\frac{14}{25}$ (4) $\frac{27}{49}$

10. A random variable X has the probability distribution

X	P(X)	X	P(X)
1	0.15	5	0.20
2	0.23	6	0.08
3	0.12	7	0.07
4	0.10	8	0.05

For the events E = {X is a prime number} and F = {X < 4}, then $P(E \cup F)$ is

(1) 0.77 (2) 0.87 (3) 0.35 (4) 0.50

11. A die is rolled so that the probability of face m (m = 1, 2, 3, 4, 5, 6) is proportional to m. The probability of getting a multiple of 3 is

(1) $\frac{3}{7}$ (2) $\frac{2}{21}$ (3) $\frac{2}{3}$ (4) $\frac{1}{7}$

12. Numbers 1, 2, 3, 4, 5, 6, 7, 8 are arranged in a random order. The probability that the digits 1, 2, 3, 4 appear as neighbours in that order, is

(1) $\frac{1}{2}$ (2) $\frac{1}{128}$ (3) $\frac{1}{256}$ (4) $\frac{1}{336}$

13. A box contains three types of seeds : 50% of type A, 20% of type B and rest of type C. It is known that 20% of A, 30% of B and 30% of C germinate. A seed is drawn randomly from the box. Find its probability to germinate.

(1) 0.25 (2) 0.50 (3) 0.80 (4) 1

14. A box contains a fair coin and a two headed coin. A coin is selected at random from the box and tossed twice. If head comes both the times, the probability that it is by the two headed coin is

(1) $\frac{1}{4}$ (2) $\frac{1}{2}$ (3) $\frac{4}{5}$ (4) $\frac{5}{8}$

15. Let X be a set containing 10 elements and P(X) be its power set. If A and B are picked up at random from P(X), with replacement, then the probability that A and B have equal number of elements, is :

(1) $\frac{(2^{10}-1)}{2^{20}}$ (2) $\frac{^{20}C_{10}}{2^{10}}$ (3) $\frac{(2^{10}-1)}{2^{10}}$ (4) $\frac{^{20}C_{10}}{2^{20}}$

16. Three critics review a book.Odds in favour of the book are 5 : 2, 4 : 3 and 3 : 4 respectively for the three critics. The probability that majority are in favour of the book is

(1) $\frac{35}{49}$ (2) $\frac{125}{343}$ (3) $\frac{164}{343}$ (4) $\frac{209}{343}$

17. An unbiased coin is tossed eight times. The probability of obtaining at least one head and at least one tail is:

(1) $\frac{255}{256}$ (2) $\frac{63}{64}$ (3) $\frac{127}{128}$ (4) $\frac{1}{2}$

18. Two uniform dice marked 1 to 6 are thrown together. The probability that the score on the two dice is at least seven is

(1) $\frac{5}{12}$ (2) $\frac{7}{12}$ (3) $\frac{3}{4}$ (4) $\frac{1}{2}$

19. If in throwing a die with six faces, probability of getting number x_i is proportional to the number then mean of this probability distribution is

(1) $\frac{15}{4}$ (2) $\frac{13}{3}$ (3) $\frac{17}{3}$ (4) $\frac{17}{4}$

20. A candidate takes three tests in succession and the probability of passing the first test is p. The probability of passing each succeeding test is p or $\frac{p}{2}$ according as he passes or fails in the preceding one. The candidate is selected, if he passes atleast two tests. The probability that the candidate is selected, is

(1) $p^2(2 - p)$ (2) $p(2 - p)$ (3) $p + p^2 + p^3$ (4) $p^2(1 - p)$

SECTION-II

(NUMERICAL VALUE TYPE QUESTIONS)

This section contains Five (5) questions. The answer to each question is **NUMERICAL VALUE** with twodigit integer and decimal upto one digit.

21. Die P has 4 red and 2 white faces whereas die Q has 2 red and 4 white faces. A fair coin is tossed. If head turns up the game continues by throwing die P. If tail turns up then die Q is to be used. If the first two throws of die resulted in red face then find probability of getting red face at the third throw ?

22. Ten coins are thrown simultaneously. If the is probability of getting at least seven heads is λ then value of 100λ?

23. Two different families A and B are blessed with equal number of children. There are 3 tickets to be distributed amongst the children of these families so that no child gets more than one ticket. If the probability that all the tickets go to the children of the family B is $\frac{1}{12}$, then the number of children in each family is :

24. Some urns contain 4 white and 6 black balls each while one urn contains 5 white and 5 black balls. One urn is chosen at random from there and 2 balls are drawn from it and both are found to be black. The probability that 5 white and 3 black balls remain in the chosen urn is $\frac{1}{7}$. The total number of urns is

25. If two different numbers are taken fromthe set {0, 1, 2, 3,....., 10}; then theprobability that their sum as well as absolute difference are both multiple of 4, is λ then find value of 55λ.

Answer Key_DPP-25

1.	**(3)**	**2.**	**(1)**	**3.**	**(3)**	**4.**	**(4)**	**5.**	**(3)**	**6.**	**(3)**	**7.**	**(2)**
8.	**(1)**	**9.**	**(4)**	**10.**	**(1)**	**11.**	**(1)**	**12.**	**(4)**	**13.**	**(1)**	**14.**	**(3)**
15.	**(4)**	**16.**	**(4)**	**17.**	**(3)**	**18.**	**(2)**	**19.**	**(2)**	**20.**	**(1)**	**21.**	**(0.6)**
22.	**(17)**	**23.**	**(5)**	**24.**	**(5)**	**25.**	**(6)**						